BACHELOR PAPER

Thesis submitted in fulfillment of the requirements for the degree of Bachelor of Science in Engineering at the University of Applied Sciences Technikum Wien - Degree Program BIF

Improving Task Efficiency in Mobile Passport Applications through Guided UI Design

By: Matthias Fichtinger
Student Number: 2310257008

Supervisor 1: Christian Lagelstorfer, BSc MSc

Hollabrunn, 19.05.2026

Declaration of Authenticity

“As author and creator of this work to hand, I confirm with my signature knowledge of the relevant copyright regulations governed by higher education acts (see Urheberrechtsgesetz/ Austrian copyright law as amended as well as the Statute on Studies Act Provisions / Examination Regulations of the UAS Technikum Wien as amended).

I hereby declare that I completed the present work independently and according to the rules currently applicable at the UAS Technikum Wien and that any ideas, whether written by others or by myself, have been fully sourced and referenced. I am aware of any consequences I may face on the part of the degree program director if there should be evidence of missing autonomy and independence or evidence of any intent to fraudulently achieve a pass mark for this work (see Statute on Studies Act Provisions / Examination Regulations of the UAS Technikum Wien as amended).

I further declare that up to this date I have not published the work to hand nor have I presented it to another examination board in the same or similar form. I affirm that the version submitted matches the version in the upload tool.”

Hollabrunn, 19.05.2026

Place, Date

Digital Signature

Kurzfassung

Mobile Anwendungen zur Erstellung biometrischer Passfotos prüfen die Einhaltung der gesetzlichen Anforderungen zuverlässig, geben Nutzern jedoch selten verständliche Hinweise darauf, was bei einem abgelehnten Foto tatsächlich korrigiert werden muss. Die Folge sind wiederholte Fehlversuche und wachsender Frust aufseiten der Nutzer*innen. In dieser Arbeit wird untersucht, wie Benutzeroberflächen mit Nutzerführung diese Kluft zwischen System und Nutzer überbrücken können.

Für die Untersuchung wurde ein plattformübergreifender Prototyp in React Native entwickelt, der über zwei Varianten verfügt: eine Baseline-Oberfläche mit minimalem Feedback nach der Aufnahme des Fotos und eine Variante mit Nutzerführung, die visuelle Overlays in Echtzeit, schrittweisen Anweisungen und kontextbezogenen Fehlermeldungen bietet. In einem kontrollierten Usability-Experiment mit 18 Teilnehmer*innen, die gleichmäßig auf beide Bedingungen aufgeteilt wurden, sind Leistungen und Wahrnehmung anhand von Bearbeitungszeit, Anzahl der Aufnahmeversuche, Erstversuchserfolg, Fehlerraten, der System Usability Scale sowie einem eigens entwickelten Likert-Fragebogen erfasst worden.

Die Ergebnisse zeigen einen deutlichen Vorteil der Variante mit Nutzerführung. Teilnehmende benötigten im Mittel 1,33 statt 4,67 Versuche, erreichten in 67 Prozent der Fälle bereits beim ersten Versuch ein konformes Foto gegenüber 11 Prozent in der Baseline-Bedingung und bewerteten das System deutlich höher (SUS 81,94 gegenüber 34,44). Die Gesamtbearbeitungszeit unterschied sich nicht signifikant, was darauf hinweist, dass geführte Nutzende pro Versuch zwar mehr Zeit investierten, insgesamt aber deutlich weniger Versuche brauchten. Aus den Befunden werden fünf konkrete Designrichtlinien für compliance-kritische mobile Anwendungen abgeleitet, die Echtzeit-Feedback, die Zerlegung von Aufgaben in Einzelschritte, visuelle Overlays für räumliche Ausrichtung, konkret umsetzbare Fehlermeldungen sowie das Design für Erstnutzende abdecken.

Schlagwörter: Benutzeroberfläche (UI), Usability, Nutzerführung, Biometrische Passbilder, Mobile Anwendungen

Abstract

Mobile applications for capturing biometric passport photographs can reliably detect whether an image meets regulatory requirements, but they rarely tell users in understandable terms what needs to be corrected when a photo is rejected. The result is repeated failed attempts, growing frustration, and poor overall user experience. This bachelor thesis investigates how guided user interface design can close this gap between what the system knows and what the user is told.

A cross-platform prototype was developed in React Native with two interface variants: a baseline interface with minimal post-capture feedback and a guided interface featuring real-time visual overlays, step-by-step instructions, and contextual error messages. In a controlled usability experiment with 18 participants, evenly split between the two conditions, performance and perception were measured through task completion time, number of attempts, first-attempt compliance, error rates, the System Usability Scale, and a custom Likert questionnaire.

The results show a clear advantage for the guided interface. Participants needed an average of 1.33 attempts compared to 4.67 in the baseline condition, achieved first-attempt compliance in 67 percent of cases against 11 percent for the baseline, and rated the system substantially higher (SUS 81.94 versus 34.44). Total task completion time did not differ significantly, indicating that guided users invested more time per attempt but required far fewer attempts overall. Based on these findings, five concrete design guidelines are derived for compliance-critical mobile applications, covering real-time feedback, task decomposition, visual overlays for spatial alignment, actionable error messaging, and designing for first-time users.

Keywords: User Interface (UI), Usability, Guided Interaction, Biometric Passport Photos, Mobile Applications

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Christian Lagelstorfer, BSc MSc, for their guidance and support throughout this thesis. I also thank all participants who took part in the usability study and contributed valuable insights. Furthermore, I would like to thank my family and friends for their continuous support and encouragement during my studies. Finally, I appreciate the support of the University of Applied Sciences Technikum Wien for providing the framework for this research.

Table of Contents

1 Introduction

1.1 Motivation & Context

Submitting a passport application increasingly requires a digital photograph that meets strict biometric standards. Yet anyone who has tried to use a mobile application for this purpose quickly realizes that achieving a compliant image is far from straightforward. Users must produce images that conform to the requirements defined by the International Civil Aviation Organization (ICAO) [1], including constraints on head position, lighting, background uniformity, and facial expression. These criteria are very precise, and even small inconsistencies can lead to rejection.

A common scenario involves a user who captures a photo, receives a generic rejection message, and has no way of knowing whether the issue was the lighting, the background, or a slightly tilted head. Without actionable guidance, users are left to repeat the process through trial-and-error, often taking multiple attempts before achieving a compliant result. This leads to longer task completion times, growing frustration, and a noticeably poor user experience.

Therefore, the interface itself plays a critical role in this context. Not just in evaluating whether a photo meets the requirements, but in actively guiding users toward a correct result. Nielsen [2] identified visibility of system status and error prevention as fundamental usability principles. Similarly, Sweller, Ayres, and Kalyuga [3] predict in their work Cognitive Load Theory that interfaces which fail to structure information effectively will increase mental effort and reduce user performance. In a mobile environment, where screen space is limited and users interact under varying conditions, these principles become especially relevant.

1.2 Problem Statement

The technical side of biometric photo validation is largely solved. Modern systems can reliably detect whether an image complies with ICAO requirements. The challenge, however, lies in the gap between what the system can detect and what the user actually understands.

In most existing mobile passport photo applications, feedback is provided only after the image has been captured. Typically, users see a simple pass or fail result. For instance, a user whose image is rejected due to a slight head tilt receives the same generic error message as a user with uneven lighting, making it impossible to determine the actual cause of failure. As a result, users are forced to guess what went wrong and try again, often repeating the same mistake multiple times before stumbling upon a compliant image.

Norman [4] traced this type of failure directly to poor or missing feedback in interactive systems. When users do not receive clear information about what went wrong, they cannot correct their behavior efficiently. Preece et al. [5] made a similar observation, noting that the absence of structured guidance pushes users toward inefficient strategies. The question, then, is not whether the system can identify errors, it clearly can, but whether the interface helps users avoid them in the first place. This thesis therefore focuses on the user-facing side of the problem: How can the interface itself support users during precision-critical tasks?

1.3 Objective of this Thesis

The objective of this thesis is to design, implement, and evaluate a guided user interface for a mobile passport photo application. The core idea is to move beyond passive, post-capture validation and instead provide active, real-time support during the photo capture process.

To this end, a functional mobile prototype was developed using React Native in combination with the Expo Framework. React Native was chosen for its cross-platform capabilities, allowing the prototype to be tested on both Android and iOS devices without maintaining separate codebases. The application includes two distinct interface variants: a baseline UI that represents a standard camera application with minimal feedback, and a guided UI that incorporates real-time visual overlays, step-by-step instructions, and contextual error messages.

Both variants were compared in a controlled usability experiment with 18 participants who were randomly assigned to one of the two conditions. The experiment evaluated differences in task completion time, number of photo attempts, first-attempt compliance, error rates, and perceived usability. A sample size of 18 was considered appropriate for this type of controlled usability study, following established recommendations in human-computer interaction (HCI) research [6] [7], and was feasible within the scope of a bachelor thesis.

1.4 Research Gap

Several foundational works in HCI [2] [4] [5] [8] agree that feedback, error prevention, and structured guidance are essential for effective user interaction. These principles are well established and have been validated across a range of application domains, including web interfaces, desktop software, and general-purpose mobile applications. However, their application in compliance-critical mobile scenarios, where users must produce outputs that meet strict, externally defined technical standards, has yet received comparatively little attention.

On the standards side, ICAO Doc 9303 [1] specifies in detail what a compliant biometric image must look like, and ISO 9241-11 [9] defines usability in terms of effectiveness, efficiency, and satisfaction. Yet neither standard addresses how users should be supported in meeting these requirements through interface design. The standards define the target but not the path toward it.

Established usability evaluation methods, including the System Usability Scale [10] and controlled experimental designs [6] [7], provide reliable tools for measuring the impact of design interventions. The System Usability Scale (SUS) was selected for this study because of its well-documented reliability across different product types [11] and its ability to produce a single composite usability score that allows direct comparison between interface conditions. Despite the availability of such methods, they have rarely been applied to evaluate guided interaction in biometric mobile applications.

To the best of the author's knowledge, no prior study has conducted a controlled comparison of guided versus non-guided interfaces specifically in the context of mobile passport photo capture. A systematic search across major HCI databases and digital libraries did not reveal comparable experimental evaluations. The present work aims to fill this gap by systematically evaluating the impact of real-time feedback, visual overlays, and step-by-step guidance on task efficiency, error rates, and compliance success.

1.5 Research Questions & Hypothesis

Based on the identified research gap, this study investigates the following research questions. The main research question emerged from the observation that existing applications can detect compliance issues reliably but fail to communicate them in a way that helps users correct their behavior.

1.5.1 Main Research Question

How do specific UI design elements, namely real-time alignment overlays, step-by-step guidance, and contextual error feedback, affect task efficiency and first-attempt compliance success in a mobile passport photo capture application?

1.5.2 Sub-Research Questions

To address this question from multiple angles, three sub-research questions were defined:

1. Does the use of real-time facial alignment overlays reduce task completion time and the number of photo attempts compared with a standard camera UI?
2. How does real-time contextual error feedback (e.g., head position, lighting warnings) influence first-attempt compliance and overall task success?
3. How do perceived usability and user satisfaction differ between guided and non-guided UI?

1.5.3 Hypothesis

Based on the theoretical background and the design rationale described above, the following hypothesis was formulated.

H1: Users interacting with a mobile passport-photo application that provides real-time guidance and contextual feedback will demonstrate higher task efficiency, fewer errors, and higher first-attempt compliance and overall task success rates within the defined time limit than users interacting with a non-guided interface.

1.6 Contribution of the Thesis

The primary contribution of this thesis is the development and empirical evaluation of a guided user interface prototype for mobile passport photo capture. The prototype was built as a cross-platform application in React Native and includes two interface variants, a baseline UI with minimal feedback and a guided UI with real-time overlays, step-by-step instructions, and contextual error messages.

The controlled usability experiment conducted with 18 participants provides empirical evidence that guided UI elements improve both objective performance metrics and subjective usability perceptions. Based on these results, a set of design guidelines is derived that extends established HCI principles to the specific context of compliance-critical mobile applications. These guidelines are intended to support designers and developers of mobile applications in which user-generated outputs must conform to predefined standards.

1.7 Structure of the Thesis

The remainder of this thesis is organized as follows:

- Chapter 2 reviews foundational HCI principles and existing work on biometric verification, identifying the research gap this thesis addresses.
- Chapter 3 details the methodology, including the experimental design, data collection, and analysis approach.
- Chapter 4 presents both the prototype produced through the design process and the empirical results of its evaluation.
- Chapter 5 discusses the findings, limitations, and design implications.
- Chapter 6 summarizes the main conclusions and suggests directions for future work.

All figures included in this thesis were created by the author unless stated otherwise.

2 State of the Art

2.1 User Interface Design and HCI Principles

How a user interface is designed has a direct impact on whether users can complete a task efficiently or end up struggling with it. This relationship has been studied extensively in human–computer interaction, and several foundational works have shaped the way interfaces are evaluated and improved today.

Nielsen [2] proposed ten usability heuristics based on a factor analysis of 249 usability problems. Among them, visibility of system status, error prevention, and recognition rather than recall are particularly relevant here: a passport photo application needs to tell users what is happening, keep them from making avoidable mistakes, and not expect them to memorize biometric requirements. Nielsen and Molich [12] showed that even informal heuristic evaluation can uncover a substantial number of usability issues early in the design process.

Sweller et al. [3] approached the problem from a cognitive perspective with their Cognitive Load Theory, distinguishing between intrinsic load from the task itself, extraneous load from poor design, and germane load that supports learning. The practical takeaway is clear: if the interface forces users to remember what went wrong or figure out compliance rules on their own, it adds unnecessary cognitive burden that competes with the task at hand.

Norman [4] approached the problem from a different angle, focusing on affordances, signifiers, and feedback as core elements of interaction design. His argument is that interfaces should make available actions obvious and respond immediately when the user does something. In a camera-based application, where there is no tactile feedback and users rely entirely on what they see on screen, this principle carries particular weight.

Tractinsky et al. [13] added another dimension by showing experimentally that users perceive visually attractive interfaces as easier to use, a reminder that first impressions can shape the entire interaction. Shneiderman et al. [8] and Preece et al. [5] rounded out the picture with practical design principles that strive for consistency, offer informative feedback, reduce memory load, and a framework for iterative, user-centered evaluation.

2.2 Real-Time Feedback and Guided Interaction

There is a meaningful difference between telling users what went wrong after they have finished a task and showing them what to adjust while they are still doing it. Most passport photo applications today follow the first approach: a user captures an image and receives a pass or fail result. If the image fails, the user starts over, often without knowing what to change. Real-time feedback takes the opposite approach; it lets users correct course before they commit to an action.

Both Norman [4] and Nielsen [2] stressed this point from different angles. Norman argued that immediate feedback is essential for users to understand the consequences of their actions. Nielsen’s heuristic of visibility of system status makes the same demand from a design perspective: the system should always keep users informed. In a passport photo scenario, this means showing a user whether their head is correctly positioned before the shutter button is pressed, not flagging it as a problem afterward.

Shneiderman et al. [8] and Preece et al. [5] described how breaking a task into smaller, sequential steps can lower the error rate, especially for first-time users. The logic is intuitive: asking a user to satisfy five requirements simultaneously is harder than walking them through one at a time. For applications that require strict compliance, this combination of real-time feedback and step-by-step guidance offers a way to support users where post-capture validation falls short.

2.3 Biometric Systems and Passport Photo Requirements

A passport photo is not just any photograph. It must meet a precise set of technical criteria defined by ICAO Doc 9303 [1], covering head position, lighting uniformity, background consistency, and facial expression. What might look fine to a human observer, a slight tilt of the head, a soft shadow under the chin, can be enough for an automated system to reject it.

ISO 9241-11 [9] adds a usability perspective: it defines usability as the extent to which a system allows users to achieve their goals with effectiveness, efficiency, and satisfaction. A passport photo application that can detect non-compliance but leaves users frustrated and confused about how to fix it is, by this definition, not usable.

On the technical side, modern biometric systems are capable enough. Face detection, landmark extraction, and image quality assessment allow them to flag even subtle deviations. The detection problem is solved. What is not solved is the feedback problem? That most applications respond to a failed check with a binary result and leave it at that. This disconnect between what the system knows and what the user is told is the gap this thesis addresses.

2.4 Usability Evaluation Methods

Measuring usability properly requires more than tracking whether users completed a task. A user who finishes in 30 seconds but found the interaction confusing has had a different experience from one who took 40 seconds but felt confident throughout. Both objective and subjective data are needed.

On the objective side, standard metrics include task completion time, number of attempts, and error counts. This is what Lazar et al. [6] described as the foundation for establishing causal links between design changes and user behavior.

On the subjective side, the System Usability Scale (SUS) developed by Brooke [10] has become one of the most widely adopted tools. It consists of ten items on a five-point Likert scale and produces a single score between 0 and 100. Bangor et al. [11] confirmed its reliability across a wide range of product types and proposed an adjective rating scale [14] mapping scores to labels: above 85 is “Excellent,” 70 to 85 is “Good,” and below 50 is “Not Acceptable.” The SUS was selected for this study because it is quick to administer, well-validated, and allows a direct comparison between the two interface conditions.

Because the SUS captures general usability but not specific dimensions like clarity of instructions or helpfulness of feedback, additional Likert-scale items are commonly used alongside it [15]. Rubin and Chisnell [7] described how to recruit participants, design tasks, and combine quantitative metrics with qualitative feedback, a mixed-method approach that is adopted in the present thesis.

2.5 UI Design and Error Prevention

When the margin for error is narrow, the interface must do more than report mistakes, it has to help users avoid them. In a passport photo application, a head tilt of a few degrees is enough to trigger rejection. Users who receive no guidance will often repeat the exact same error, simply because they have no information to work with. Nielsen [2] listed error prevention as one of his core heuristics, and Norman [4] argued that good design should make errors hard to commit in the first place. Shneiderman et al. [8] and Sweller et al. [3] added that when errors do occur, feedback must be specific, not a generic “Image rejected.”

In mobile contexts, these challenges are amplified by changing lighting, inconsistent backgrounds, and limited screen space. For passport photo applications, providing real-time overlays, step-by-step instructions, and contextual error messages during capture offers a practical path to reducing errors before they happen. The concepts and methods presented in this chapter form the theoretical foundation for the experimental evaluation described in the following chapter.

3 Methodology

3.1 Research Design

The research design for this study is a controlled usability experiment comparing two interface conditions: a baseline UI with minimal feedback and a guided UI with real-time overlays, step-by-step instructions, and contextual error messages.

A between-subjects design was used, meaning that each participant was randomly assigned to one of the two conditions, and interacted with only that UI version. A within-subjects design, in which participants would test both versions, was also considered but ultimately rejected because of the risk of learning effects. A user who has already completed the task once would approach the second attempt with knowledge from the previous attempt, making it impossible to isolate the effect of interface from the effect of prior experience. The primary objective of the experiment was to compare user performance and perceived usability between the two UI variants under controlled conditions.

3.2 Controlled Usability Experiment

3.2.1 Experimental Design

The study follows a between-subjects design with two conditions: baseline UI and guided UI. Participants were randomly assigned to one of the two conditions to ensure comparability and reduce selection bias. Each participant interacted with only one interface variant.

3.2.2 Participants

A total of 18 participants were recruited for the experiment, resulting in 9 participants per condition. No prior experience with biometric photo applications was required. In fact, the absence of such experience was desirable, as it ensures the results reflect typical first-time user behavior rather than learned strategies from previous use.

3.2.3 Task

Each participant was given a single task: capture a passport-style photograph that meets the predefined biometric requirements. No additional instructions about the specific requirements were provided. The task was designed to simulate a realistic usage scenario in which users must independently figure out how to achieve compliance.

3.2.4 Procedure

The task began when the participant pressed the “Start Test” button, which activated the camera interface, and ended when a compliant image was successfully captured and the participant pressed the “End Test” button. Participants could take as many photo attempts as needed. A maximum time limit of five minutes was imposed. If no compliant image was achieved within this window, the task was recorded as unsuccessful.

3.3 Data Collection

3.3.1 Objective Measures

The application automatically recorded the following performance metrics during task execution: start and end timestamps, number of photo attempts, first-attempt compliance (binary), task completion status (binary), total number of errors, warnings, and success messages, the distribution of feedback messages across validation categories, and a complete attempt history including timestamps and associated feedback for each attempt.

3.3.2 Subjective Measures

Subjective user experience was assessed using the System Usability Scale (SUS) and a custom five-item Likert-scale questionnaire. The Likert items covered clarity of instructions, helpfulness of feedback, perceived task efficiency, confidence that the photo would be accepted, and overall satisfaction. Additionally, participants were given the opportunity to provide open-ended qualitative feedback about their experience.

3.3.3 Data Preprocessing

Prior to analysis, raw data were preprocessed to ensure consistency. Task completion time was derived from start and end timestamps. SUS scores were computed according to the standard scoring method, and Likert-scale responses were aggregated into mean scores per participant. Several derived metrics were also calculated: errors per attempt, warnings per attempt, and success rate.

3.3.4 Measurement Definitions

To ensure consistency and reproducibility, all variables are defined as follows:

- Start timestamp: Timestamp set the moment the participant pressed “Start Test” button
- End timestamp: Timestamp set the moment the participant pressed “End Test” button or time limit of 5 minutes was reached
- Number of photo attempts: Total number of images captured, and compliance checked by the participant during the task (minimum 1)
- First-attempt compliance: Indicates whether the first compliance check met compliance requirements (1 = yes, 0 = no)
- Task completion status: Indicates whether a compliant image was achieved within the time limit (1 = success, 0 = failure)
- Total number of errors, warnings, and success messages: For each the total number will be collected whenever the compliance check is triggered
- Distribution of feedback messages: For each feedback message the total number will be collected whenever the compliance check is triggered
- Attempt history: For each attempt errors, warnings, and success messages are collected whenever the compliance check is triggered as well as a timestamp at which the attempt was created
- SUS Score: Calculated according to the standard SUS scoring method (range: 0 – 100)
- Likert-scale scores: Numerical responses (1-5) for each questionnaire item
- Open-ended feedback: Qualitative user comments collected after task completion
- Errors per attempt: Total number of errors divided by number of attempts
- Warnings per attempt: Total number of warnings divided by number of attempts
- Success rate: Ratio of success messages to total feedback messages

The structure of the processed dataset is illustrated in Appendix A (Figure 14).

3.4 Data Analysis

3.4.1 Overview of Analysis Strategy

The analysis is structured around the three sub-research questions (SRQ1-SRQ3), with additional analyses to explore patterns in user interaction behavior. Due to the relatively small sample size ($n = 18$, with 9 participants per condition), non-parametric methods are used as the primary statistical approach, since they do not require the assumption of normally distributed data. Where normality can reasonably be assumed, parametric tests are applied. All analyses were conducted in Python using a significance level of $\alpha = 0.05$. Effect sizes are reported alongside significance tests to assess the practical magnitude of observed differences, which is an important complement given the limited sample.

3.4.2 Quantitative Analysis

Task Efficiency (SRQ1). Task efficiency is assessed through two metrics: task completion time and number of photo attempts. Both variables are continuous and not assumed to be normally distributed. The Mann-Whitney U test is used to compare the two groups, as it is suitable for independent samples with non-normal distributions and small sample sizes.

Compliance and Task Success (SRQ2). First-attempt compliance and task completion status are both binary outcomes (success or failure). The Chi-square test of independence is applied to compare proportions between conditions. In cases where expected cell frequencies fall below conventional thresholds, Fisher's exact test is used instead, as it provides exact p-values without relying on asymptotic approximations.

Usability and User Perception (SRQ3). SUS scores and Likert-scale ratings are compared between conditions. Where the data appear approximately normally distributed, independent samples t-tests are applied. Otherwise, the Mann-Whitney U test serves as a non-parametric alternative. These measures capture how users perceived the interaction rather than how they performed objectively.

3.4.3 Additional Analysis

Error and Feedback Analysis. Beyond the primary research questions, error and feedback metrics are examined in more detail. Total error counts, warning counts, errors per attempt, and the success rate are all compared between conditions using the Mann-Whitney U test. Normalizing errors by the number of attempts is particularly important here, as it accounts for the fact that participants who take more attempts will naturally accumulate more errors in absolute terms.

Correlation Analysis. Spearman rank correlation is used to explore whether the number of attempts is associated with the total error count, and whether task completion time correlates with the number of attempts. These correlations help clarify whether different aspects of user performance are linked or independent of each other.

Attempt-Level and Learning Analysis. Attempt-level data are analyzed to determine whether users improved their performance across successive attempts. If baseline participants show a decreasing error trend over time while guided participants start with low error rates and stay there, this would suggest that the guided UI reduces the need for learning through repeated failure. Errors per attempt across sequential attempts are compared between conditions using the Mann-Whitney U test.

Error Type Analysis. Feedback messages are broken down by validation category, including face detection, lighting, background, positioning, and expression, to identify which aspects of the task benefit most from guided UI elements. The frequency of errors, warnings, and successes per category is compared between conditions. This analysis reveals whether guidance has a uniform effect across all validation steps or whether its impact is concentrated in specific areas.

3.4.4 Effect Size Reporting

In addition to statistical significance, effect sizes are reported for all key comparisons. For Mann-Whitney U tests, the rank-biserial correlation r is used. For t-tests, Cohen’s d is reported. For Chi-square and Fisher’s exact tests, odds ratios or Cramer’s V are provided where applicable. Given the small sample size of this study, effect sizes are particularly important: a statistically significant result with a small effect may have limited practical relevance, while a non-significant result with a large effect may still be meaningful and warrant further investigation with a larger sample.

3.4.5 Interpretation Strategy

Results are interpreted primarily based on statistical significance at $\alpha = 0.05$. However, significance alone is not treated as the sole criterion for drawing conclusions. Effect sizes are considered alongside p-values to assess whether observed differences are large enough to matter in practice. For non-significant results, the direction and magnitude of the effect are still discussed if they suggest a meaningful trend that could reach significance with a larger sample.

3.4.6 Qualitative Analysis

Open-ended participant responses were analyzed using a thematic analysis approach, following three steps:

1. Familiarization: Responses were read multiple times to develop an overall sense of the data and to identify recurring patterns.
2. Coding: Relevant statements were assigned descriptive labels such as “lack of guidance”, “trial-and-error”, “clear instructions”, “ease of use”, and “frustration”. Coding was performed manually by the researcher. To support the identification of recurring patterns and ensure consistency, AI-assisted tool (ChatGPT) was used to suggest initial code groupings. These suggestions were critically reviewed and refined manually.
3. Theme Development: The identified codes were grouped into higher-level themes based on conceptual similarity. Four themes emerged:
    a. Clarity of Instructions
    b. User Confidence
    c. Perceived Efficiency
    d. User Experience

The qualitative findings are used to complement and contextualize the quantitative results, providing a richer picture of how participants experience the two interface conditions.

3.5 Ethics and Data Protection

All participants provided informed consent before taking part in the experiment. Participation was voluntary, and participants were free to withdraw at any time without providing a reason.

All collected data were anonymized immediately after collection. No personal identifying information was retained beyond what was necessary to conduct the analysis. The study complies with applicable data protection regulations, including the General Data Protection Regulation (GDPR). All data, analysis scripts, and source code are provided in the appendix and supplementary materials (see Appendix E) to ensure transparency and reproducibility.

4 Results

This chapter presents two types of results. Section 4.1 describes the prototype produced through the design process, including its system architecture, the two interface variants that constitute the experimental conditions, and the underlying validation logic. Sections 4.2 onward report the empirical findings from the controlled usability experiment evaluating this prototype.

4.1 Prototype

4.1.1 System Architecture

A functional prototype was developed as a cross-platform mobile application for Android and iOS using React Native in combination with the Expo Framework. The application enables users to capture passport-style photographs and receive feedback on biometric compliance. Biometric validation was supported by a locally executed backend service implemented in Python. This backend simulates compliance checks using predefined rule-based logic. The decoupling of validation from the frontend was a deliberate architectural choice: it allows both interface variants, described in the following section, to share an identical validation pipeline, so that any observed difference between conditions can be attributed to the UI itself. Local execution further eliminates network variability as a source of timing artifacts. The objective was not commercial-grade accuracy but consistency and reproducibility across all sessions.

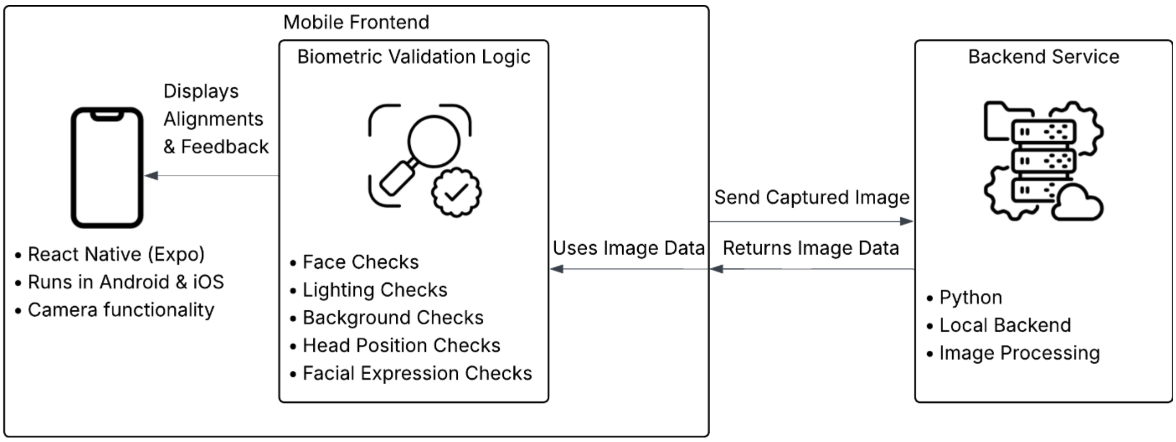

Figure 1: System Architecture of Mobile Prototype

4.1.2 User Interface Design

Two user interface variants resulted from the design process. Both share the same underlying system architecture and validation logic. The only difference between them is the level of guidance provided to the user during the capture process. Each design decision is grounded in HCI principles drawn from the literature reviewed in Chapter 2.

Baseline UI. The baseline interface implements a standard camera application with minimal feedback, as shown in Figure 3. It includes basic camera controls such as camera switching and flash functionality, as well as the ability to retake images. After capturing a photo, the system indicates whether the image meets compliance requirements but provides no further detail about what went wrong, as visible in Figure 2. This minimal design was intentional: the baseline was designed as a methodological control condition, deliberately restricted in its feedback to isolate the absence of in-task guidance as the experimental manipulation.

Figure 3: Baseline UI Camera Overlay

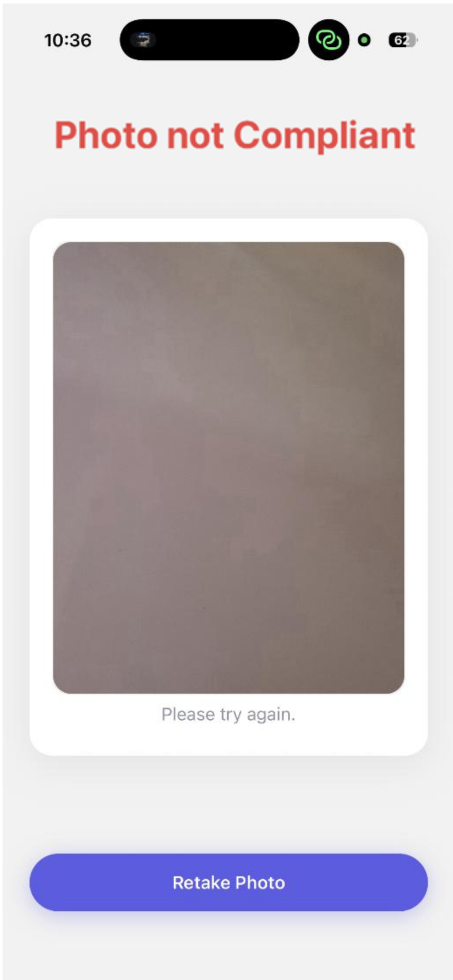

Figure 2: Baseline UI Photo Not Compliant

Guided UI. The guided interface extends the baseline with three guidance elements grounded in HCI literature: step-by-step instructions for each validation step reducing extraneous cognitive load [3], real-time visual overlays for head position, tilt, or size operationalizing visibility of system status [2], and contextual feedback messages that explain detected issues or warnings as informative feedback [4]. At each step, the system clearly explains to the user what needs to be done, as illustrated in Figure 4. Once a step is successfully completed, a confirmation message appears and the next step is triggered automatically. After all steps have been completed, the system informs the user that it is the right time to capture the photo. If the final image still does not meet the required criteria, the guided interface provides specific feedback explaining which requirement was not met, as shown in Figure 5. This is the key difference between the two variants: the guided interface offers continuous support throughout the process, while the baseline interface provides feedback only at the end.

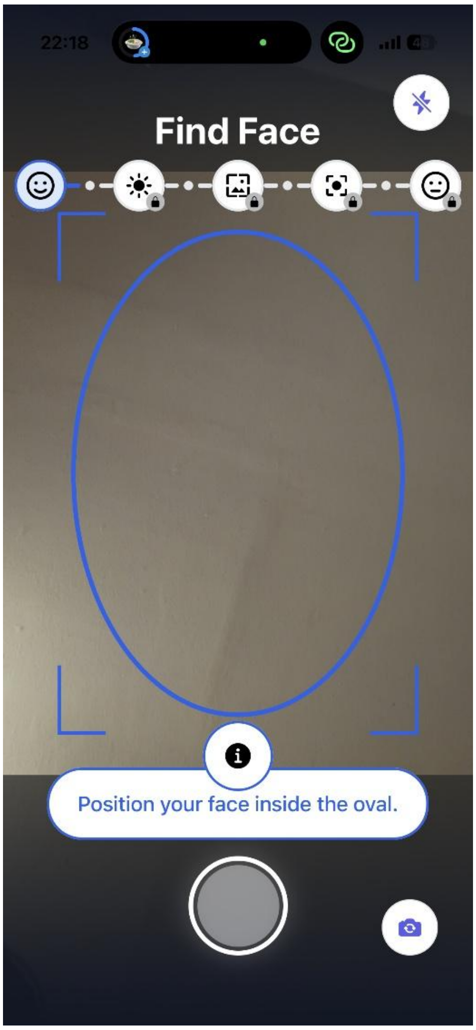

Figure 4: Guided UI Camera Overlay

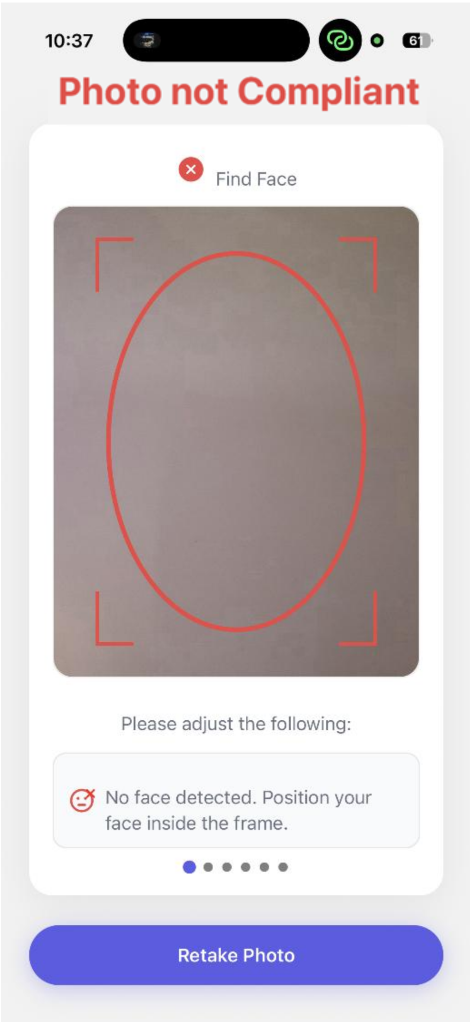

Figure 5: Guided UI Photo Not Compliant

4.1.3 Biometric Validation Logic

The biometric compliance checks are implemented using simplified rule-based logic rather than complex machine learning models. This was a deliberate choice: the goal is not to replicate the accuracy of production systems, but to ensure that the validation behavior is consistent and reproducible across all participants. In a controlled experiment, what matters is that every user encounters the same checks under the same conditions, not that the system can handle edge cases it would face in deployment. Rule-based logic is also deterministic and inspectable, which supports reproducibility in a way that a learned model, whose decisions can drift across versions or training runs, cannot. The backend evaluates captured images based on the following criteria, selected to cover the major ICAO categories [1] while distinguishing between user-controllable adjustments and environment-dependent factors. This distinction becomes methodologically relevant in the step-level analysis (Section 4.9), which examines whether real-time guidance is equally effective across all categories or whether its impact concentrates on those the user can directly act on:

1. Face Detection: Verifying that exactly one face is present.
2. Lighting Conditions: Analyzing overall brightness and detecting backlighting.
3. Background Uniformity: Evaluating consistency within acceptable ranges.
4. Head Position: Estimating rotation, size, and verifying centering using facial landmarks.
5. Facial Expression: Detecting whether face expression is neutral.

These checks form the basis for both real-time feedback in the guided UI and the final compliance evaluation in both interface conditions.

4.2 Overview of Collected Data

A total of 18 participants were included in the experiment, with 9 assigned to each condition. The dataset comprises both objective performance measures (task completion time, number of attempts, error and warning counts) and subjective usability measures (System Usability Scale and Likert-scale ratings). Non-parametric tests were applied for variables not assumed to be normally distributed; parametric tests were used where appropriate. A significance level of $\alpha = 0.05$ was applied throughout. Effect sizes are reported for all comparisons to support interpretation given the small sample size. An overview of the dataset structure is provided in Appendix A (Figure 14).

4.3 Task Efficiency (SRQ1)

4.3.1 Task Completion Time

Task completion time did not differ significantly between conditions ($U = 50.0$, $p = 0.427$, $r = 0.20$). Guided UI participants finished in 57.80 seconds on average, compared to 65.55 seconds in the baseline condition. While the guided condition shows a descriptively lower median (Figure 6), variability between participants was similar across both groups. The small effect size suggests this difference has little practical relevance. A complete overview of all statistical test results is provided in Appendix B (Table 1).

Figure 6: Boxplot - Task Completion Time for Baseline and Guided UI

4.3.2 Number of Attempts

The clearest performance difference emerged in the number of attempts. Guided UI participants needed only 1.33 attempts on average to achieve a compliant image, compared to 4.67 in the baseline condition (U = 75.0, p = 0.002, r = 0.72). As Figure 7 shows, the guided condition exhibits both a lower median and much less spread. The large effect size confirms that this is not just statistically significant but practically meaningful. Detailed descriptive statistics are provided in Appendix C (Table 2).

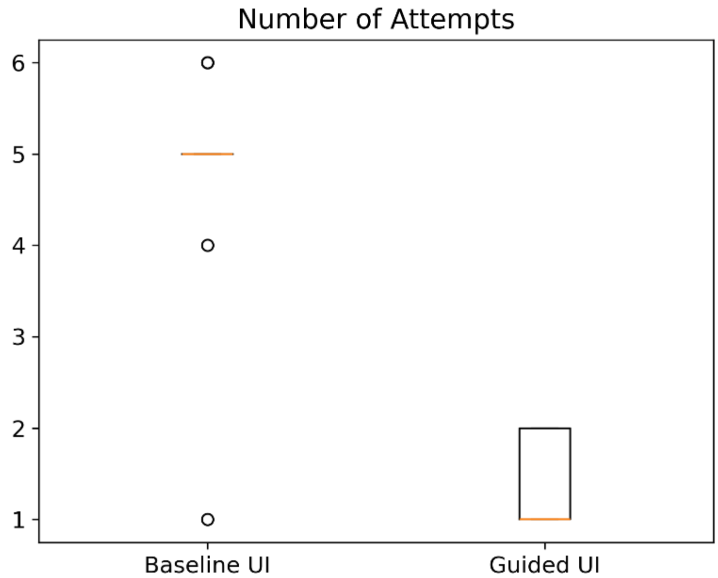

Figure 7: Boxplot - Number of Attempts for Baseline and Guided UI

4.4 Compliance and Task Success (SRQ2)

4.4.1 First-Attempt Compliance

Six out of nine guided UI participants achieved compliance on their very first attempt. In the baseline condition, only one out of nine did. Fisher's exact test confirmed this association ($p = 0.050$, $OR = 16.0$), meaning guided users were sixteen times more likely to succeed on the first try. Figure 8 illustrates the contrast.

4.4.2 Task Completion Success

Overall task completion success did not differ between conditions (Fisher's exact, $p = 1.000$). Every participant in both groups managed to produce a compliant image within the five-minute time limit. This ceiling effect limits what can be concluded from this metric alone. Both interfaces allowed users to reach the goal, but the guided UI got them there with substantially fewer attempts (see Section 4.3.2).

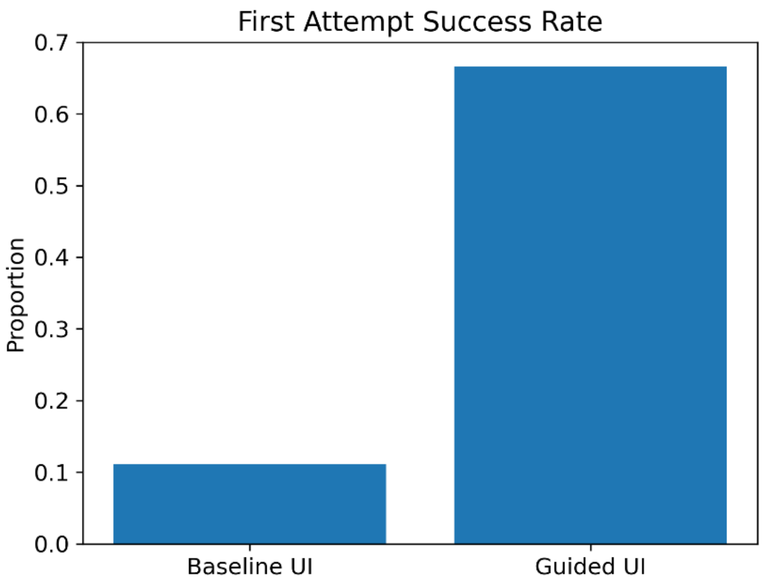

Figure 8: Bar Chart - First-Attempt Success Rate

4.5 Usability and User Perception (SRQ3)

4.5.1 System Usability Scale (SUS)

The SUS scores produced the starkest contrast between conditions ($t = -11.72$, $p < 0.001$, $d = 5.53$). The guided UI averaged 81.94, placing it in the “Good” to “Excellent” range on Bangor et al.’s adjective scale [14]. The baseline UI scored 34.44, which falls into the “Not Acceptable” range. Figure 9 (left) illustrates this gap: the guided UI consistently achieves higher scores with noticeably lower variability. An effect size of $d = 5.53$ is exceptionally large, even by the standards of controlled HCI experiments (see Appendix C for summary statistics).

4.5.2 Likert-Scale Evaluation

Likert-scale ratings followed the same pattern ($t = -11.20$, $p < 0.001$, $d = 5.28$). Guided UI participants rated every single item higher than baseline participants: clarity of instructions, helpfulness of feedback, perceived task efficiency, confidence in acceptance, and overall satisfaction. The guided UI averaged 4.22 out of 5, compared to 2.00 for the baseline. Figure 9 (right) shows this difference across all five dimensions.

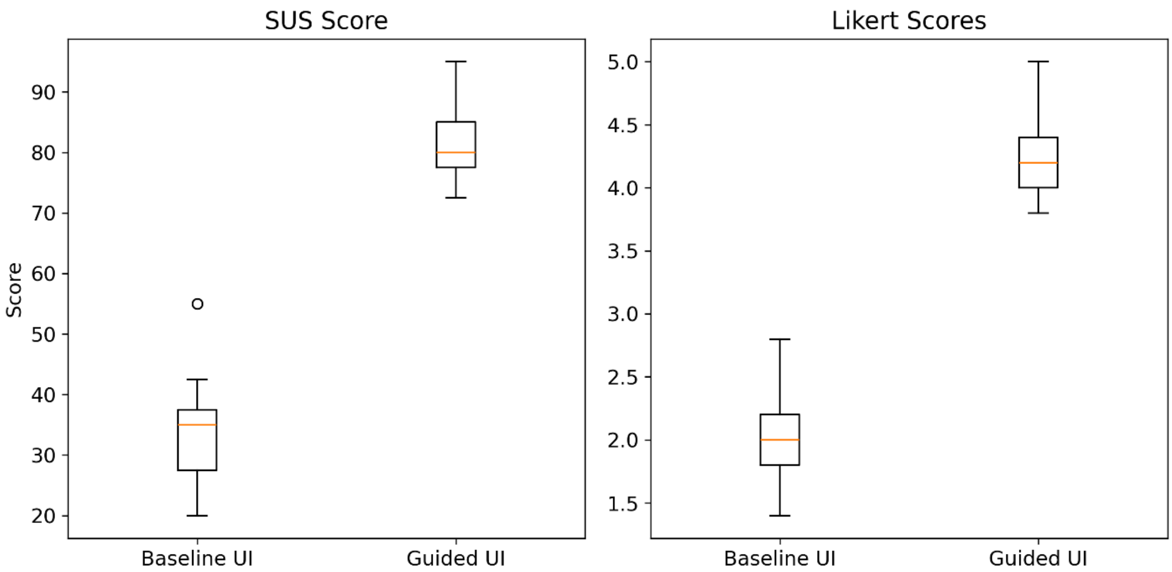

Figure 9: Boxplots - SUS Score (left) and Likert-Scale Ratings (right) for Baseline and Guided UI

4.6 Error and Feedback Analysis

Across all error-related metrics, the guided UI performed significantly better. Total errors dropped from 6.33 in the baseline to 0.44 in the guided condition ($U = 75.0$, $p = 0.002$, $r = 0.72$). Even when normalized per attempt, the difference remained clear: errors per attempt fell from 1.24 to 0.22 ($U = 74.5$, $p = 0.002$, $r = 0.71$), indicating that baseline users were not just making more attempts but also more mistakes within each attempt. Warnings showed a similar reduction, from 2.67 to 0.33 ($U = 72.0$, $p = 0.004$, $r = 0.66$). The success rate, defined as the proportion of positive compliance feedback relative to total feedback, was also significantly higher in the guided condition (0.89 vs. 0.66, $U = 2.0$, $p < 0.001$, $r = 0.80$). The consistency of these effects across different metrics reinforces the conclusion that the guided interface did not merely reduce the number of attempts but also improved the quality of each individual attempt. Figure 10 shows 3 Graphs illustrating these differences. A full summary of error-related metrics is provided in Appendix C (Table 2).

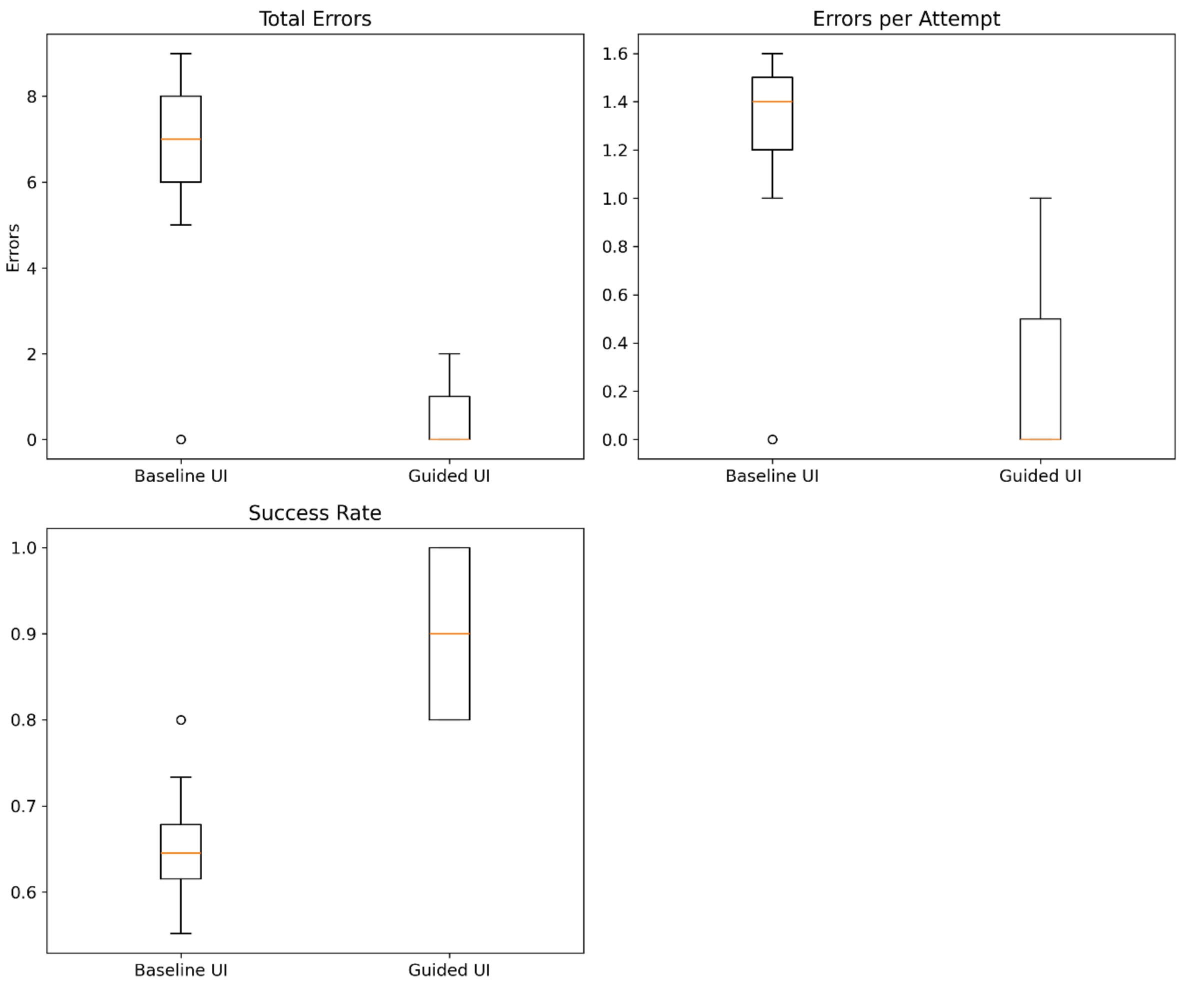

Figure 10: Boxplots - Error and Feedback Metrics: Total Errors (top left), Errors per Attempt (top right), and Success Rate (bottom) for Baseline and Guided UI

4.7 Correlation Analysis

4.7.1 Attempts and Errors

A Spearman rank correlation revealed a strong positive relationship between the number of attempts and total errors ($r_s = 0.972$, $p < 0.001$). As Figure 11 shows, participants who needed more attempts accumulated substantially more errors. This relationship held consistently across both UI conditions.

4.7.2 Task Completion Time and Attempts

No significant correlation was found between task completion time and the number of attempts ($r_s = 0.386$, $p = 0.114$). Participants who took more attempts did not necessarily need more total time. One likely explanation is that baseline participants captured images quickly but had to repeat the process, while guided participants spent more time per attempt but needed fewer of them.

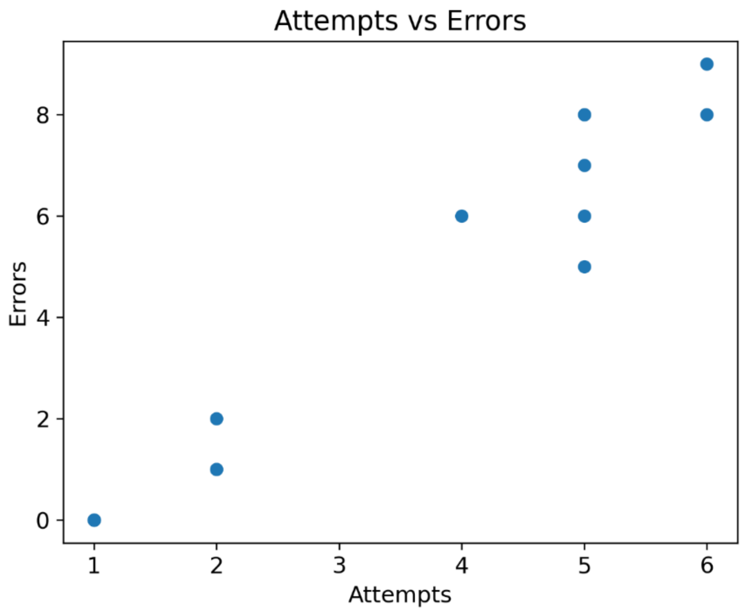

Figure 11: Scatter Plot of Attempts vs Total Errors

4.8 Learning Effects

Attempt-level error data were compared across sequential attempts using the Mann-Whitney U test. A significant difference was found ($U = 397.0$, $p = 0.002$). As Figure 12 shows, the baseline UI exhibits a clear downward trend in errors across attempts: participants started with high error rates and gradually improved, suggesting a trial-and-error learning process. The guided UI, by contrast, maintained consistently low error levels from the first attempt onward. This pattern indicates that the guided interface compressed the learning process into the first interaction, eliminating the need for users to improve through repeated failure.

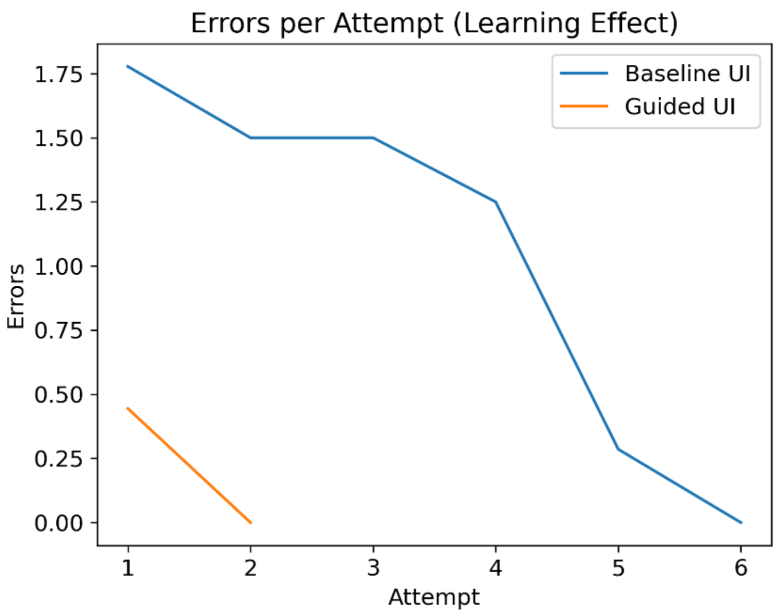

Figure 12: Line Chart - Learning Curve showing Errors per Attempt Across Sequential Attempts

4.9 Step-Level Analysis

Error occurrences were broken down by validation category and compared between conditions using the Mann-Whitney U test. For the “Adjust Face Position” step, a significant reduction in errors was observed in the guided condition ($U = 24.0$, $p = 0.014$). This step involves positioning the head within the correct region of the frame, which is directly supported by the real-time alignment overlay. For the “Adjust Background” step, no significant difference was found ($U = 7.0$, $p = 0.301$), which is expected given that background quality depends primarily on the physical environment rather than on anything the interface can guide the user to change in real time. Figure 13 illustrates these differences. The results suggest that guided UI elements are most effective for validation criteria that involve user-controllable behavior.

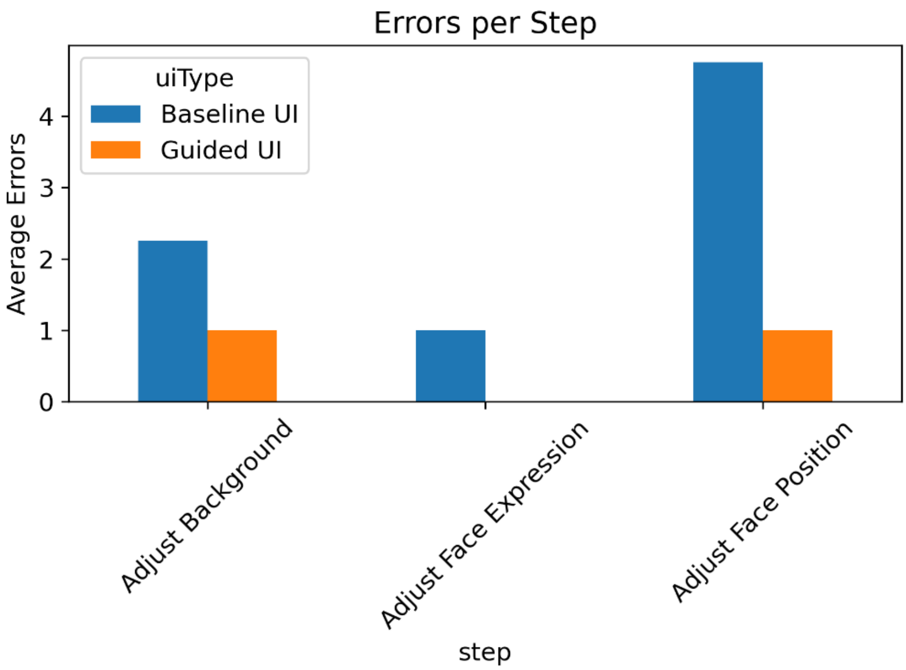

Figure 13: Bar Chart - Average Errors per Validation Step for Baseline and Guided UI

4.10 Qualitative Results

Open-ended feedback was collected from 17 of 18 participants (one did not provide a response). The responses were coded and grouped into themes using thematic analysis as described in Section 3.4.6.

Baseline participants consistently described a lack of guidance and frustration with unclear feedback. Representative responses include:

- P01: “Instructions could have been more clear. I didn’t know what to do really.”
- P03: “No clear instructions on how to take photo. Try and error was required.”
- P07: “No guidance at all. Frustrating if photo gets rejected multiple times but I don’t know why.”
- P15: “Very frustrating. I didn’t know what to fix after errors.”
- P17: “Needed multiple tries. No clear feedback why it failed.”

Guided participants, by contrast, described the experience in markedly different terms:

- P02: “Guidance through the photo taking process was fluent and easy to understand.”
- P04: “Well designed guidance. Overlays were very helpful.”
- P08: “Good guidance, fixed quickly after one mistake.”
- P16: “Very intuitive. Fixed the issue quickly.”
- P18: “Perfect guidance. Very easy to complete.”

Four themes emerged from the coding process. The most prominent was Clarity of Instructions: baseline users reported missing or unclear instructions, while guided users described the step-by-step process as clear and easy to follow. Closely related was User Confidence, where baseline participants expressed hesitation and uncertainty while guided participants reported feeling in control throughout. The third theme, Perceived Efficiency, captured the contrast between the trial-and-error approach described by baseline users and the structured, fast process reported by guided users. Finally, the broader theme of User Experience reflected the overall emotional tone: frustration and confusion on the baseline side, smoothness and intuitiveness on the guided side. A complete overview of all responses with assigned codes and themes is provided in Appendix D (Table 3).

4.11 Summary of Results

Table 1 below provides a summary of all statistical comparisons conducted in this chapter.

Metric	Test	Statistic	p-value	Effect Size	Result
Task Completion Time	MWU	50.00	0.427	r = 0.20	Not significant
Number of Attempts	MWU	75.00	0.002	r = 0.72	Significant
First-Attempt Compliance	Fisher’s	–	0.050	OR = 16.0	Significant
Task Completion Success	Fisher’s	–	1.000	–	Not significant
SUS Score	t-test	−11.72	< 0.001	d = 5.53	Significant
Likert Score	t-test	−11.20	< 0.001	d = 5.28	Significant
Total Errors	MWU	75.00	0.002	r = 0.72	Significant
Errors per Attempt	MWU	74.5	0.002	r = 0.71	Significant
Warnings	MWU	72.0	0.004	r = 0.66	Significant
Success Rate	MWU	2.00	< 0.001	r = 0.80	Significant
Attempts vs Errors	Spearman	0.972	< 0.001	–	Significant
Time vs Attempts	Spearman	0.386	0.114	–	Not significant
Attempt-Level Errors	MWU	397.00	0.002	–	Significant
Errors: Adjust Background	MWU	7.00	0.301	–	Not significant
Errors: Adjust Face Position	MWU	24.00	0.014	–	Significant

Table 1: Summary of statistical comparisons between baseline and guided UI (MWU = Mann-Whitney U, Fisher's = Fisher's Exact)

Statistically significant differences were found for the number of attempts, first-attempt compliance, SUS scores, Likert-scale ratings, total errors, errors per attempt, warnings, and success rate. No significant differences were found for task completion time and overall task completion success. The guided UI reduced errors across most validation steps, with the strongest effect observed for face positioning. Qualitative feedback consistently distinguished between the two conditions, with baseline participants reporting frustration and lack of guidance, and guided participants reporting clarity and confidence.

5 Discussion

The previous chapter presented the quantitative and qualitative results of the usability experiment. This chapter interprets those findings, places them in the context of existing HCI research, and discusses what they mean for the design of compliance-critical mobile applications. The chapter also addresses the limitations of the study and concludes with a set of design guidelines derived from the empirical evidence.

5.1 Interpretation of Results

The most notable outcome of this experiment is not any single metric, but the consistency of the pattern across nearly all measures. Participants in the guided UI condition needed fewer attempts, made fewer errors, achieved higher first-attempt compliance, and rated the system as substantially more usable. Taken together, these results provide strong support for the stated hypothesis (H1).

The reduction in the number of photo attempts is perhaps the clearest indicator of the guided UI's effectiveness. Guided participants needed an average of 1.33 attempts to produce a compliant image, compared to 4.67 in the baseline condition ($p = 0.002$, $r = 0.72$). In practical terms, most guided users got it right on the first or second try, while baseline users captured and resubmitted their photo four to six times before succeeding. The real-time overlay and the step-by-step instructions appear to have given users enough information to position themselves correctly before pressing the shutter button, something the baseline interface simply did not offer.

What makes this finding interesting is that it did not translate into a significant difference in overall task completion time ($p = 0.427$, $r = 0.20$). At first glance, this seems counterintuitive: if guided users needed fewer attempts, one would expect them to finish faster. A closer look suggests an explanation. Guided participants spent more time per individual attempt because the interface walked them through multiple steps before allowing the photo to be captured. Baseline participants, by contrast, captured images quickly but then had to repeat the process multiple times. The result was roughly similar total durations in both conditions, but through very different interaction patterns. The non-significant correlation between task completion time and the number of attempts (r_s = 0.386, p = 0.114) supports this interpretation.

First-attempt compliance tells a similar story from a different angle. In the guided condition, 6 out of 9 participants achieved a compliant image on their very first try. In the baseline condition, only 1 out of 9 were able to do so ($p = 0.050$, $OR = 16$). This difference is difficult to dismiss, even with a small sample size. It suggests that the guided interface did not merely help users recover from errors faster. It prevented most of the errors from occurring in the first place.
Overall task completion success, on the other hand, did not differ between conditions. Regardless of the interface condition, every participant eventually achieved a compliant photo within the five-minute time limit. This ceiling effect limits what can be concluded from this metric alone. However, it does highlight an important distinction: the baseline interface is not fundamentally broken. Users can still reach the goal. The problem is how they get there, through frustration, guesswork, and repeated failure rather than through structured support by the application.

The subjective measures reinforce this distinction. The SUS score for the guided UI was 81.94 placing it in the “Good” to “Excellent” range according to Bangor et al.’s adjective scale [14]. The baseline UI scored 34.44, firmly in the “Not Acceptable” range. The gap between these scores ($d = 5.53$) is unusually large, even by the standards of controlled HCI experiments. Likert-scale ratings followed the same pattern ($d = 5.28$), with guided participants reporting higher clarity, better feedback, greater confidence, and more satisfaction across all five items.

The error analysis adds further depth. Guided users encountered a mean of 0.44 errors across the entire task, compared to 6.33 in the baseline condition ($p = 0.002$). Even when normalized per attempt, the guided UI still resulted in substantially fewer errors (0.22 vs 1.24, $p = 0.002$). This means that baseline users were not just making more attempts. They also made more mistakes within each attempt. Warnings showed a similar reduction (0.33 vs 2.67, $p = 0.004$), and the overall success rate was significantly higher in the guided condition (0.89 vs 0.66, $p < 0.001$).

The learning curve comparison revealed that baseline participants gradually reduced their errors across successive attempts, suggesting they were learning through trial-and-error. Guided participants, by contrast, started with low error rates and stayed there. This pattern indicates that the guided interface compressed the learning process into the first interaction. Users did not need multiple failures to understand what was expected of them.

The step-level analysis showed that guidance was most effective when users had direct control over the outcome. Face positioning errors dropped significantly in the guided condition ($U = 24.0$, $p = 0.014$), which makes sense given that the overlay provided a direct visual reference for correct head placement. Background errors, however, did not differ significantly ($p = 0.301$). All participants were tested under the same controlled conditions and could reposition themselves if needed, so the lack of difference likely reflects that background adjustments require the user to physically relocate rather than simply adjust their posture, making this type of correction less responsive to on-screen guidance.

5.2 Comparison with Existing Research

The results align closely with several principles in HCI, though they also extend these principles into a domain where they have not been empirically tested before.

The core argument of this thesis, that real-time feedback during task execution outperforms post-task feedback, is consistent with what Nielsen [2] and Norman [4] have long advocated. Nielsen's heuristic of visibility of system status predicts that systems which keep users informed about what is happening will produce fewer errors and higher satisfaction. Norman's work on feedback and affordances makes a similar point from a cognitive perspective: users need to see the effect of their actions immediately in order to learn and correct course. The data from this experiment confirm both predictions. Baseline participants, who received feedback only after capture, made significantly more errors and reported significantly lower satisfaction than guided participants, who received continuous status information during the task.

Sweller, Ayres, and Kalyuga's Cognitive Load Theory [3] offers another lens through which to interpret the results. In the baseline condition, users had to remember the requirements, capture an image, interpret a generic error message, recall what they did differently, and try again. Each failed attempt added to the mental burden. The guided interface reduced this load by externalizing the relevant information. Users did not have to remember what to do, because the interface told them in real time. The significant reduction in errors per attempt (0.22 vs. 1.24) is exactly the kind of outcome Sweller et al. [3] would predict when extraneous cognitive load is minimized.

The step-by-step structure of the guided UI connects directly to the recommendations of Shneiderman et al. [8], who described how sequential task decomposition can reduce errors and support unfamiliar users. Preece et al. [5] made a related observation about the role of structured guidance in reducing the learning curve. Both predictions were confirmed: guided users performed well from the first attempt, while baseline users required multiple iterations to achieve similar results.

The qualitative data provide perhaps the most vivid illustration of these theoretical principles in practice. Baseline participants used words like "frustrating," "trial and error," and "no guidance at all," language that maps directly onto what Norman [4] described as the consequences of poor feedback design. Guided participants described the experience as "fluent," "intuitive," and "easy to understand," suggesting that the interface successfully implemented the principles these authors have been advocating for decades.

It is worth noting, however, that the magnitude of the effects observed in this study is larger than what is typically reported in comparable HCI experiments. The Cohen's d values for SUS

(5.53) and Likert scores (5.28) are exceptionally high. One possible explanation is that the baseline condition in this study was deliberately minimal. It provided almost no guidance at all. In a comparison against a more refined but still non-guided interface, the differences might be smaller. This should be kept in mind when interpreting the practical significance of the results.

5.3 Implications for UI Design

Several practical implications follow from these results, particularly for developers working on mobile applications where users must produce outputs that meet strict external requirements.

The most direct implication is that post-capture validation alone is insufficient. When the system only tells the user whether their photo passed or failed, without showing what went wrong or how to fix it, the result is repeated failure and frustration. The data makes a clear case for shifting feedback upstream: instead of checking the output after the fact, the interface should guide users toward a correct result during the process. This is not a new idea in HCI theory, but the experimental evidence presented demonstrates its practical impact in a specific, real-world application context.

A second implication concerns the scope of what guidance can achieve. The step-level analysis showed that guidance was highly effective for face positioning, where the real-time overlay gave users a direct visual reference they could respond to by slightly tilting their head or adjusting their posture. For background quality, the guided UI did provide feedback on whether the background met the requirements, but this did not lead to a significant reduction in errors. The likely reason is that correcting a background issue requires the user to physically relocate rather than make a small postural adjustment, which is a more demanding response to on-screen feedback. This suggests that simply telling users about a problem is not always enough. When the required correction involves a larger physical action, developers may need to intervene earlier in the process, for example by running a background check before the capture workflow begins and prompting the user to find a suitable location before they start.

A third implication relates to the learning curve. The experiment showed that baseline users gradually improved across attempts. They were learning, but through repeated failure. Guided users did not need this learning phase because the interface provided the necessary information upfront. This finding is especially relevant for applications that are used infrequently. A user who captures a passport photo once every five or ten years cannot be expected to remember the requirements from a previous session. The interface must assume that every user is a first-time user, and it must be designed accordingly.

Finally, the qualitative results suggest that guidance affects not only performance but also the emotional quality of the interaction. Baseline participants reported frustration and uncertainty, while guided participants reported confidence and ease. Even in cases where both groups eventually succeeded, the subjective experience was fundamentally different. For applications where user trust and willingness to return are important, such as government services or regulated processes, this difference in experience should not be underestimated.

5.4 Limitations

Several limitations should be considered when interpreting the results of this study. The most obvious limitation is the sample size. With 9 participants per condition, the study is small by conventional standards. Although statistically significant differences were found for most metrics and the observed sizes were consistently large, small samples increase the risk that individual outliers disproportionately influenced the results. A larger sample would provide more robust estimates and allow for subgroup analyses, for instance by age, technical proficiency, or prior experience with photo applications. Future studies should aim for a larger and more diverse participant pool to strengthen the external validity of these findings.

The biometric validation logic used in the prototype is a simplified, rule-based implementation rather than a production-grade computer vision system. This was a deliberate choice: for a controlled experiment, consistent and reproducible system behavior is more important than detection accuracy. Every participant encountered the same validation rules under the same conditions, which ensures that observed differences can be attributed to the interface rather than to variations in system performance. That said, the specific numbers reported in this study, such as error rates and compliance thresholds, are tied to the rule-based logic used in the prototype. A commercial application using more advanced detection algorithms may flag different issues or apply stricter criteria, which could shift the absolute performance numbers even if the relative advantage of guided over non-guided interfaces remains.

The experiment was conducted under controlled conditions in a single room with consistent lighting and background, though participants were free to move around or leave the room if they chose to. This setup was necessary to ensure that performance differences between conditions could be attributed to the interface rather than to environmental factors. However, it also means the results primarily reflect a controlled scenario. In everyday use, people capture passport photos in varied settings with different lighting, backgrounds, and device angles. Whether the guided UI provides the same level of improvement under less predictable conditions was not tested in this study and remains a question for future research.

The between-subjects experiment design eliminates learning effect between conditions but introduces the possibility that the two groups differed in ways not captured by random assignment. With only 9 participants per group, even small differences in prior experience could influence the outcome. A within-subjects design, in which each participant would test both interfaces, would have allowed each person to serve as their own control. However, once a participant has completed the task with one interface, they inevitably learn what a compliant photo requires. That prior knowledge would carry over into the second condition, making it impossible to tell whether their improved performance was caused by the interface or by what they had already learned during the first round.

Lastly, all participants used the application for the first time. The results therefore reflect typical first-time user behavior, which is arguably the most relevant scenario for a passport photo application. Nevertheless, it is likely that the performance gap between guided and non-guided interfaces narrows with repeated use as users internalize the requirements. A long-term study examining how the effect of guidance evolves over multiple sessions would complement the present findings.

5.5 Design Guidelines for Task-Critical Mobile Applications

Based on the findings of this study and the theoretical foundations reviewed in Chapter 2, the following design guidelines are proposed. They are intended for designers and developers of mobile applications in which users must produce outputs that conform to strict technical or regulatory requirements, whether in biometric verification, medical imaging, document scanning, or similar domains.

The first guideline is to ***provide real-time feedback during task execution, not only after completion***. The experiment demonstrated that users who received continuous feedback made fewer errors and achieved compliance more quickly than users who were informed only after the fact. This aligns with Nielsen's [2] heuristic of visibility of system status. In practice, this means integrating live indicators such as color-coded status bars, position markers, and contextual feedback prompts that update as the user adjusts their behavior.

The second guideline is to ***deconstruct complex tasks into sequential, manageable steps***. Rather than confronting the user with all requirements simultaneously, the interface should address one criterion at a time. This reduces the cognitive load placed on users [3] and allows them to focus their attention on a single adjustment. The guided UI in this study implemented this approach by presenting validation steps in sequence, and the results suggest it was effective.

The third guideline is to ***use visual overlays for spatial alignment tasks***. The step-level analysis showed that face positioning errors were significantly reduced by the real-time alignment overlay, while background-related errors were not. This shows that visual guides are most effective when users can directly act on feedback, for example by tilting their head or moving their device.

The fourth guideline is to ***replace generic error messages with specific, actionable feedback***. Baseline participants in this study consistently reported that they did not know what to fix after receiving rejection. A message stating that the photo is “not compliant” provides no direction for improvement. Instead, the system should identify specific issues and, where possible, suggest concrete corrective action.

The fifth guideline is to ***design for first-time users and assume no prior knowledge***. The learning curve analysis revealed that guided users performed well from the first attempt, while baseline users needed several attempts to reach comparable performance. Applications that are used infrequently, such as passport photo tools or identity verification flows, cannot rely on users building up experience over time.

These guidelines do not replace existing HCI principles, they add to them in a specific application context that has received limited empirical attention. The foundational recommendations of Nielsen [2], Norman [4], Shneiderman et al. [8], and Preece et al. [5] remain as relevant as ever. What this study adds is concrete evidence that these principles, when translated into real-time guidance, visual overlays, and step-by-step instructions, produce measurable improvements in both performance and user experience in compliance-critical mobile tasks.

6 Conclusion

6.1 Summary

This study investigated whether guided UI design can improve task efficiency and compliance success in mobile passport photo applications. The motivation was a concrete gap in existing applications, because they detect non-compliance accurately and reliably, but they rarely communicate what needs to be corrected in a way that users act on it. To test whether real-time guidance can close this gap, a React Native prototype with two interface conditions was developed and compared in a controlled usability experiment with 18 participants.

The results support the stated hypothesis across nearly every measure. Guided participants averaged 1.33 attempts compared to 4.67 for the baseline, achieved first-attempt compliance in 67 percent of cases against 11 percent, and rated the interface much higher on both the SUS (81.94 vs 34.44) as well as the Likert questionnaire. Error rates and warnings dropped significantly in the guided variant, and the qualitative feedback pointed in the same direction. Task completion time and overall completion success showed no significant difference. Guided users spent more time per individual attempt, and all participants eventually produced a compliant image within the five-minute time limit.

Beyond confirming the hypothesis, the work produced two contributions. As the step-level analysis revealed, guidance is most effective for user-controlled factors such as head positioning, and less effective when it requires the user to make a larger physical correction, like relocating for the background condition. Building on these findings, five design guidelines were derived for compliance-critical mobile applications, covering real-time feedback, task decomposition, visual overlays, actionable error messaging, and designing for first-time users. The guidelines extend existing HCI principles into a context that had received little empirical attention.

The main limitations of this study are the small sample size, the simplified validation logic, and the controlled experimental setting. These factors constrain how directly the absolute numbers transfer to commercial applications, but the relative advantage of the guided interface over the baseline appears robust across all measured metrics.

6.2 Future Work

Several directions for future work follow from the limitations discussed in this thesis. A larger and more diverse sample would make the findings more generalizable and allow for comparisons between subgroups, for example by age, technical proficiency, or language background. Replacing the rule-based validation logic with a production-grade computer vision system would show whether the observed effects still hold when detection is stricter and less predictable.

A long-term study would complement the present work by examining how the effect of guidance changes across multiple sessions, which is something the current design cannot answer. Field testing in real-world settings, with varied lighting and environments, would show whether the same improvements hold outside of a controlled scenario.

Finally, the design guidelines proposed here could be tested in related domains such as document scanning, identity verification, or medical imaging, where users also need to produce results that meet strict requirements. Repeating the study in these areas would show how well these guidelines apply and where they need to be adjusted to each context.

Bibliography

[1] International Civil Aviation Organization (ICAO), Doc 9303: Machine Readable Travel Documents – Part 9: Deployment of Biometric Identification and Electronic Storage of Data in eMRTDs, 8th ed. Montreal, Canada: ICAO, 2021. [Online]. Available: https://www.icao.int/publications/doc-series/doc-9303 [Accessed: May 6, 2026].

[2] J. Nielsen, "Enhancing the explanatory power of usability heuristics," in Proc. ACM CHI'94 Conf. Human Factors in Computing Systems, Boston, MA, Apr. 1994, pp. 152–158. doi: 10.1145/191666.191729.

[3] J. Sweller, P. Ayres, and S. Kalyuga, Cognitive Load Theory. New York, NY: Springer, 2011. doi: 10.1007/978-1-4419-8126-4.

[4] D. A. Norman, The Design of Everyday Things, Revised and Expanded ed. New York, NY: Basic Books, 2013. ISBN 978-0-465-05065-9.

[5] H. Sharp, J. Preece, and Y. Rogers, Interaction Design: Beyond Human-Computer Interaction, 5th ed. Chichester, UK: Wiley, 2019. ISBN 978-1-119-54725-9.

[6] J. Lazar, J. H. Feng, and H. Hochheiser, Research Methods in Human-Computer Interaction, 2nd ed. Cambridge, MA: Morgan Kaufmann, 2017. ISBN 978-0-12-805390-4.

[7] J. Rubin and D. Chisnell, Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests, 2nd ed. Indianapolis, IN: Wiley, 2008. ISBN 978-0-470-18548-3.

[8] B. Shneiderman, C. Plaisant, M. Cohen, S. Jacobs, N. Elmqvist, and N. Diakopoulos, Designing the User Interface: Strategies for Effective Human-Computer Interaction, 6th ed. Boston, MA: Pearson, 2016. ISBN 978-0-13-438038-4.

[9] International Organization for Standardization, ISO 9241-11:2018 – Ergonomics of Human-System Interaction – Part 11: Usability: Definitions and Concepts. Geneva, Switzerland: ISO, 2018. [Online]. Available: https://www.iso.org/standard/63500.html [Accessed: May 6, 2026].

[10] J. Brooke, "SUS: A 'quick and dirty' usability scale," in Usability Evaluation in Industry, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London, UK: Taylor and Francis, 1996, pp. 189–194, ISBN 978-0-7484-0460-5.

[11] A. Bangor, P. Kortum, and J. Miller, "An empirical evaluation of the System Usability Scale," Int. J. Human-Computer Interaction, vol. 24, no. 6, pp. 574–594, 2008. doi: 10.1080/10447310802205776.

[12] J. Nielsen and R. Molich, "Heuristic evaluation of user interfaces," in Proc. ACM CHI'90 Conf. Human Factors in Computing Systems, Seattle, WA, Apr. 1990, pp. 249–256. doi: 10.1145/97243.97281.

[13] N. Tractinsky, A. S. Katz, and D. Ikar, "What is beautiful is usable," Interacting with Computers, vol. 13, no. 2, pp. 127–145, Dec. 2000. doi: 10.1016/S0953-5438(00)00031-X.

[14] A. Bangor, P. Kortum, and J. Miller, "Determining what individual SUS scores mean: Adding an adjective rating scale," J. Usability Studies, vol. 4, no. 3, pp. 114–123, May 2009. [Online]. Available: https://uxpajournal.org/wp-content/uploads/sites/7/pdf/JUS_Bangor_May2009.pdf [Accessed: May 6, 2026].

[15] J. Sauro and J. R. Lewis, Quantifying the User Experience: Practical Statistics for User Research, 2nd ed. Cambridge, MA: Morgan Kaufmann, 2016. ISBN 978-0-12-802308-2.

List of Figures

List of Tables

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CLT	Cognitive Load Theory
GDPR	General Data Protection Regulations
HCI	Human-Computer Interaction
ICAO	International Civil Aviation Organization
ISO	International Organization for Standardization
mHealth	Mobile Health
MWU	Mann-Whitney U
SRQ	Sub-Research Question
SUS	System Usability Scale
UI	User Interface
UX	User Experience

Documentation table of AI-based tools

AI-based tools	Intended use	Prompt, source, page, paragraph...
ChatGPT (4.0)	Support for qualitative data analysis (initial code grouping and theme suggestion; results reviewed and refined by the author)	“Group the following qualitative responses into codes and themes”, Section 3.4.6, Section 4.10
ChatGPT (4.0)	Grammar, spelling, and style check across the thesis (identification of typos, grammatical errors, and unclear phrasing; suggestions reviewed and accepted or rejected by the author)	“Check the following text for grammar and clarity”, Throughout thesis

A: Processed Dataset Structure

The following excerpt (Figure 14) illustrates the structure of the processed dataset in JSON format. For clarity, non-essential fields have been omitted. The complete dataset, including all participants, is available in the GitHub repository "PassportPhotoCapture-Backend" in both JSON and CSV formats under the name “processed_data.json” or “processed_data.csv” (see Appendix E).

Example of Participants Data


```json
{
  "participantId": "P04",
  "uiType": "optimized",
  "attempts": 1,
  "firstAttemptComplianceSuccess": 1,
  "taskSuccess": 1,

  "errorCount": 0,
  "warningCount": 0,
  "successesCount": 5,

  "taskTime": 115.34,
  "SUS": 72.5,
  "Likert": 4.0,

  "errorsPerAttempt": 0.0,
  "successRate": 1.0,

  "successes": {
    "Find Face": { "6": 1 },
    "Adjust Background": { "3": 1 }
  },

  "attemptHistory": [
    {
      "attempt": 1,
      "results": [
        { "step": "Find Face", "hint": "Success" },
        { "step": "Adjust Background", "hint": "Success" }
      ]
    }
  ],

  "open_feedback": "Well designed guidance..."
}
```


Figure 14: Structure of the processed dataset in JSON format (excerpt)

B: Statistical Test Results

The table below presents all statistical tests performed on the collected data, including the applied test, test statistic, p-value, result, effect size, and effect type. The full dataset is available in the GitHub repository “PassportPhotoCapture-Backend” in both JSON and CSV format (see Appendix E). MWU = Mann-Whitney U.

Metric	Test	Statistic	p-value	Result	Effect Size	Effect Type
Task Time	MWU	50.00	0.427	Not significant	0.198	r
Attempts	MWU	75.00	0.002	Significant	0.718	r
First-Attempt Compliance	Fisher’s Exact	–	0.050	Significant	16.0	OR
Task Success	Chi-square	0.00	1.000	Not significant	–	–
SUS Score	t-test	−11.72	< 0.001	Significant	5.527	Cohen’s d
Likert Score	t-test	−11.20	< 0.001	Significant	5.279	Cohen’s d
Total Errors	MWU	75.00	0.002	Significant	0.718	r
Warnings	MWU	72.00	0.004	Significant	0.656	r
Errors per Attempt	MWU	74.50	0.002	Significant	0.708	r
Success Rate	MWU	2.00	< 0.001	Significant	0.801	r
Attempts vs. Errors	Spearman	0.97	< 0.001	Significant	0.972	r_s
Time vs. Attempts	Spearman	0.39	0.114	Not significant	0.386	r_s
Attempt-Level Errors	MWU	397.00	0.002	Significant	0.411	r
Errors: Adjust Background	MWU	7.00	0.301	Not significant	0.387	r
Errors: Adjust Face Position	MWU	24.00	0.014	Significant	0.739	r

Table 2: Summary of statistical tests, including applied methods, test statistics, p-value, effect size, and effect type

C: Summary Statistics

The table below presents descriptive statistics for each metric by UI condition. The full dataset is available in the GitHub repository “PassportPhotoCapture-Backend” in both JSON and CSV format (see Appendix E).

Metric	UI Type	Mean	Median	SD	Q1	Q3	IQR	Min	Max	n
Task Time (s)	Baseline	65.55	68.96	39.57	52.07	80.95	28.88	15.84	108.83	9
Task Time (s)	Guided	57.80	54.15	23.46	45.10	57.67	12.57	35.44	115.34	9
Attempts	Baseline	4.67	5.00	1.50	5.00	5.00	0.00	1.00	6.00	9
Attempts	Guided	1.33	1.00	0.50	1.00	2.00	1.00	1.00	2.00	9
Errors	Baseline	6.33	7.00	2.69	6.00	8.00	2.00	0.00	9.00	9
Errors	Guided	0.44	0.00	0.73	0.00	1.00	1.00	0.00	2.00	9
Warnings	Baseline	2.67	2.00	2.35	1.00	3.00	2.00	0.00	8.00	9
Warnings	Guided	0.33	0.00	0.50	0.00	1.00	1.00	0.00	1.00	9
Errors / Attempt	Baseline	1.24	1.40	0.50	1.20	1.50	0.30	0.00	1.60	9
Errors / Attempt	Guided	0.22	0.00	0.36	0.00	0.50	0.50	0.00	1.00	9
Success Rate	Baseline	0.66	0.65	0.08	0.63	0.68	0.06	0.55	0.80	9
Success Rate	Guided	0.89	0.90	0.09	0.80	1.00	0.20	0.80	1.00	9
SUS Score	Baseline	34.44	35.00	10.29	27.50	37.50	10.00	20.00	55.00	9
SUS Score	Guided	81.94	80.00	6.47	77.50	85.00	7.50	72.50	95.00	9
Likert Score	Baseline	2.00	2.00	0.48	1.80	2.20	0.40	1.40	2.80	9
Likert Score	Guided	4.22	4.20	0.35	4.00	4.40	0.40	3.80	5.00	9
First-Attempt Compl.	Baseline	0.11	0.00	0.33	0.00	0.00	0.00	0.00	1.00	9
First-Attempt Compl.	Guided	0.67	1.00	0.50	0.00	1.00	1.00	0.00	1.00	9
Task Success	Baseline	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	9
Task Success	Guided	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	9

Table 3: Descriptive statistics for baseline and guided UI conditions

D: Qualitative Analysis

The table below presents the qualitative analysis of participant responses. A total of 17 responses were collected, one participant (P12) did not provide feedback. The full dataset is available in the GitHub repository “PassportPhotoCapture-Backend” in Excel format (see Appendix E).

ID	UI Type	Comment	Codes	Theme
P01	Baseline	Instructions could have been more clear. I didn’t know what to do really. There were no instructions.	lack_of_guidance, clarity	Clarity of Instructions
P02	Guided	Guidance through the photo taking process was fluent and easy to understand. Alignments made it clear what to do. Clear instructions.	clarity, ease_of_use, guidance	Clarity of Instructions
P03	Baseline	No clear instructions on how to take photo. Try and error was required until i had compliant photo.	clarity, trial_and_error	Clarity of Instructions
P04	Guided	Well designed guidance. Overlays were very helpful. And instructions clear.	clarity, guidance	Clarity of Instructions
P05	Baseline	No instructions at all. Only if photo compliant or not. Multiple try’s were needed until compliant photo was achieved.	lack_of_guidance, trial_and_error	Clarity of Instructions
P06	Guided	Helpful guidance.	guidance	User Confidence
P07	Baseline	No guidance at all. Frustrating if photo gets rejected multiple times but I don’t know why.	lack_of_guidance, frustration, guidance	Clarity of Instructions
P08	Guided	Good guidance, fixed quickly after one mistake.	efficiency, guidance	User Confidence
P09	Baseline	Without guidance very hard to achieve a compliant photo.	guidance	User Confidence
P10	Guided	Minor adjustment needed but overall very easy.	ease_of_use	User Confidence
P11	Guided	Very smooth, only slight lighting adjustment needed.	–	General Usability
P12	Baseline	(No comment)	–	–
P13	Guided	Clear instructions, well designed.	clarity	Clarity of Instructions
P14	Baseline	No instructions. It’s hard to achieve a compliant photo with no help.	lack_of_guidance	Clarity of Instructions
P15	Baseline	Very frustrating. I didn’t know what to fix after errors.	frustration	User Experience
P16	Guided	Very intuitive. Fixed the issue quickly.	ease_of_use, efficiency	User Confidence
P17	Baseline	Needed multiple tries. No clear feedback why it failed.	clarity	Clarity of Instructions
P18	Guided	Perfect guidance. Very easy to complete.	ease_of_use, guidance	User Confidence

Table 4: Overview of qualitative participant responses, including assigned codes and identified themes

E: Supplementary Materials

All Source code datasets, and analysis scripts used in this study are available in the following repositories:

Frontend (Mobile Application):

- https://github.com/CaZZe-dv/PassportPhotoCaptureApp-Frontend

Backend (Validation, Analysis, and Plots):

- https://github.com/CaZZe-dv/PassportPhotoCapture-Backend

The repositories include:

- Mobile application prototype (React Native)
- Backend validation logic, Analysis scripts, Datasets, and Plot creation (Python)